AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	TX facility TX_way_1432637062 -- station KSAT, America/Chicago
Committed pair	OSM 1432637061 -> 1432637062 OSM way 1432637061 / OSM way 1432637062
Facade gap	83.6 m between the two halls
Weather record	43,809 real hourly records from KSAT
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3698 C at 150 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2022-03-25 (crossing)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 11 of 24 hours with 2 mode changes. The reactive incumbent took 12 hours -- 1 more -- but declared 1 unsafe hour against the agent's 0. 1 hour was safe but still ran chillers, because the schedule could not afford them.

Agent	11 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) 1 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	10.643	21.0	10.001	1.416
01	FREE	yes	-	10.228	21.0	9.440	1.284
02	FREE	yes	-	10.057	21.0	9.441	1.060
03	FREE	yes	-	8.564	21.0	6.670	0.860
04	FREE	yes	-	9.229	21.0	7.221	1.116
05	FREE	yes	-	8.954	21.0	6.671	1.038
06	FREE	yes	-	7.292	21.0	6.670	0.934
07	FREE	yes	-	7.521	21.0	6.111	0.871

08	FREE	yes	-	11.289	21.0	11.111	1.465
09	FREE	yes	-	16.282	21.0	17.013	2.535
10	FREE	yes	-	18.526	21.0	20.927	2.503
11	mechanical	no	dry-bulb	22.694	21.0	25.930	1.734
12	mechanical	no	dry-bulb	25.631	21.0	27.473	1.390
13	mechanical	no	dry-bulb	27.882	21.0	29.212	1.355
14	mechanical	no	dry-bulb	31.026	21.0	30.367	1.436
15	mechanical	no	dry-bulb	31.301	21.0	30.367	1.325
16	mechanical	no	dry-bulb	31.479	21.0	30.348	1.304
17	mechanical	no	dry-bulb	31.425	21.0	30.326	1.455
18	mechanical	no	dry-bulb	29.869	21.0	28.648	1.340
19	mechanical	no	dry-bulb	27.791	21.0	25.148	1.702
20	mechanical	no	dry-bulb	25.132	21.0	20.081	1.652
21	mechanical	no	dry-bulb	24.323	21.0	19.612	1.844
22	mechanical	no	dry-bulb	22.400	21.0	19.252	1.593
23	mechanical	yes	switch budget	19.070	21.0	17.581	1.333

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00   FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.643 C, which is 10.357 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.416 C, and the margin is measured, not chosen: 1.070 C of group-conditional forecast error for this hour of day, plus 0.3460 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

01:00   FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.228 C, which is 10.772 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.284 C, and the margin is measured, not chosen: 0.926 C of group-conditional forecast error for this hour of day, plus 0.3582 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

02:00   FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.057 C, which is 10.943 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.060 C, and the margin is measured, not chosen: 0.768 C of group-conditional forecast error for this hour of day, plus 0.2916 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

03:00   FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.564 C, which is 12.436 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.860 C, and the margin is measured, not chosen: 0.632 C of group-conditional forecast error for this hour of day, plus 0.2289 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

04:00   FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.229 C, which is 11.771 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.116 C, and the margin is measured, not chosen: 0.767 C of group-conditional forecast error for this hour of day, plus 0.3496 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.954 C, which is 12.046 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.038 C, and the margin is measured, not chosen: 0.688 C of group-conditional forecast error for this hour of day, plus 0.3496 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.292 C, which is 13.708 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.934 C, and the margin is measured, not chosen: 0.598 C of group-conditional forecast error for this hour of day, plus 0.3362 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.521 C, which is 13.479 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.871 C, and the margin is measured, not chosen: 0.570 C of group-conditional forecast error for this hour of day, plus 0.3005 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.289 C, which is 9.711 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.465 C, and the margin is measured, not chosen: 1.161 C of group-conditional forecast error for this hour of day, plus 0.3038 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.282 C, which is 4.718 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.535 C, and the margin is measured, not chosen: 2.156 C of group-conditional forecast error for this hour of day, plus 0.3788 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.526 C, which is 2.474 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.503 C, and the margin is measured, not chosen: 2.124 C of group-conditional forecast error for this hour of day, plus 0.3788 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.694 C against a 21.0 C limit, so it fails by 1.694 C. It would take a limit 1.694 C higher, or a bound 1.694 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.631 C against a 21.0 C limit, so it fails by 4.631 C. It would take a limit 4.631 C higher, or a bound 4.631 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.882 C against a 21.0 C limit, so it fails by 6.882 C. It would take a limit 6.882 C higher, or a bound 6.882 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 31.026 C against a 21.0 C limit, so it fails by 10.026 C. It would take a limit 10.026 C higher, or a bound 10.026 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 31.301 C against a 21.0 C limit, so it fails by 10.301 C. It would take a limit 10.301 C higher, or a bound 10.301 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 31.478 C against a 21.0 C limit, so it fails by 10.478 C. It would take a limit 10.478 C higher, or a bound 10.478 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 31.425 C against a 21.0 C limit, so it fails by 10.425 C. It would take a limit 10.425 C higher, or a bound 10.425 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 29.869 C against a 21.0 C limit, so it fails by 8.869 C. It would take a limit 8.869 C higher, or a bound 8.869 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.792 C against a 21.0 C limit, so it fails by 6.792 C. It would take a limit 6.792 C higher, or a bound 6.792 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.132 C against a 21.0 C limit, so it fails by 4.132 C. It would take a limit 4.132 C higher, or a bound 4.132 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.323 C against a 21.0 C limit, so it fails by 3.323 C. It would take a limit 3.323 C higher, or a bound 3.323 C tighter, to change this hour.

22:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.400 C against a 21.0 C limit, so it fails by 1.400 C. It would take a limit 1.400 C higher, or a bound 1.400 C tighter, to change this hour.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.070 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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